

Cross-Domain Transfer and Efficient Adaptation of Vision-Language Models for Medical Visual Question Answering

Andres Cabrera
George Mason University

Mann Mihir Desai
George Mason University

Abstract

Pretrained vision-language models (VLMs) achieve strong performance on general visual question answering yet degrade significantly on specialized medical imagery, where visual concepts such as fractures, lesions, and organ morphology differ fundamentally from web-scraped training data. We investigate two questions: (1) how well does domain-specific fine-tuning of a VLM on one medical subdomain (e.g., radiology) transfer to other subdomains (e.g., pathology), and (2) what is the minimum parameter-efficient adaptation needed to achieve clinically useful accuracy? We fine-tune LLaVA-1.5-7B [2] using LoRA [4] on three medical VQA datasets: VQA-RAD [1], PathVQA [7], and SLAKE [8]-and systematically evaluate cross-domain transfer via a 3×3 transfer matrix. A LoRA rank sweep ($r \in \{4, 8, 16, 32, 64, 128\}$) identifies the Pareto-optimal trade-off between adapter size and accuracy. Fine-tuning with only 0.6% of model parameters yields +12.96 percentage-point overall accuracy gain on VQA-RAD, radiology-to-radiology transfer generalizes asymmetrically, and pathology constitutes a visually isolated domain that does not benefit from radiology fine-tuning.

1. Introduction

Medical Visual Question Answering (Med-VQA) sits at the intersection of computer vision and natural language processing, requiring a model to jointly interpret a medical image-such as a radiology scan or pathology slide-and produce a free-form answer to a clinical question. Decision support tools that can answer questions such as “Is there pleural effusion?” or “What organ is shown?” could alleviate the growing burden on radiologists, who routinely read hundreds of scans per day [9].

Despite impressive progress in general-purpose VLMs, deploying them in medical settings remains challenging for two reasons. First, training data scarcity is acute: VQA-RAD, the primary radiology VQA benchmark, contains only 3,515 question-answer pairs across 315 images, orders of magnitude smaller than natural-image datasets. Second, medical imaging subdomains are visually heterogeneous.

Radiology (CT, MRI, X-ray) captures organ-level anatomy at centimeter scale in grayscale, while pathology (histology microscopy) reveals cellular structures at micron scale with color-stained tissue. A model fine-tuned on one subdomain may therefore be entirely blind to the other.

Recent work on generative VLMs, including LLaVA-Med [3] and Med-Flamingo, has shown that instruction-tuning large models on medical Q&A pairs produces strong results. However, full fine-tuning updates all ~ 7 billion parameters per dataset, making controlled cross-domain experiments prohibitively expensive. Parameter-efficient fine-tuning methods such as LoRA [4] and QLoRA [5] offer a practical alternative: by injecting low-rank adapter matrices into transformer layers, they reduce trainable parameters by two to three orders of magnitude while maintaining competitive performance.

We leverage this efficiency to conduct two studies that, to our knowledge, have not been reported together in the literature. **Contribution 1-Cross-Domain Transfer Study:** We produce a 3×3 transfer matrix by training one LoRA adapter per dataset and evaluating every adapter on all three test sets, directly answering whether a radiology-tuned model can handle pathology without retraining. **Contribution 2-LoRA Efficiency Analysis:** We sweep LoRA rank from 4 to 128 on VQA-RAD and identify the Pareto-optimal configuration, mapping adapter size to accuracy and quantifying the minimum compute required for acceptable clinical VQA performance.

2. Related Work

Discriminative classifiers. Early Med-VQA systems such as MEVF (2019) and MMQ (2021) pair a CNN image encoder with a Bi-LSTM question encoder and feed their fused representations into a classification head over a fixed answer vocabulary. While effective on a single dataset, these models cannot generate answers unseen at training time and require per-dataset retraining-properties that preclude cross-domain transfer studies [9]. This paradigm serves as the foundational baseline our work improves upon.

Self-supervised medical pretraining. M3AE (2022) and MUMC (2023) pretrain multi-modal masked autoencoders on millions of paired medical image–report examples before fine-tuning task-specific heads. Although they achieve strong in-domain results, the pretraining cost is substantial, outputs remain discriminative, and the per-task head design complicates fair domain transfer analysis.

Generative vision-language models. LLaVA [2] connects a frozen CLIP visual encoder [10] to a Vicuna-7B language model via a trainable linear projector, then instruction-tunes the LLM on GPT-4-generated multimodal conversations. LLaVA-Med [3] adapts this recipe with biomedical image-text pairs and demonstrates state-of-the-art generative Med-VQA performance. Our work builds on LLaVA-1.5-7B and extends it to a controlled cross-domain setting enabled by LoRA’s parameter efficiency. BLIP-2 [6] similarly bridges vision and language through a lightweight Q-Former; we follow the simpler LLaVA projection scheme for reproducibility.

3. Method

3.1. Base Architecture

Our backbone is **LLaVA-1.5-7B** (`llava-hf/llava-1.5-7b-hf`), comprising three components that are kept frozen throughout all experiments: **Vision encoder.** A CLIP ViT-L/14 model processes each input image resized to 336×336 pixels and outputs a sequence of visual feature vectors (≈ 0.6 GB).

Multimodal projector. A single linear layer maps CLIP features into the LLM’s embedding space, bridging the vision and language modalities.

Language model. Vicuna-7B, an instruction-tuned variant of LLaMA-2, serves as the causal language model backbone (≈ 13 GB in `bfloat16`).

Only the LoRA adapter weights are trainable.

3.2. LoRA Adapter Design

We follow the formulation of Hu et al. [4], modifying each target weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ as:

$$h = Wx + \frac{\alpha}{r}(BA)x, \quad (1)$$

where $A \in \mathbb{R}^{r \times d_{\text{in}}}$ and $B \in \mathbb{R}^{d_{\text{out}} \times r}$ are the learned low-rank matrices, r is the rank, and α is a scaling hyperparameter fixed to $2r$ in all experiments. We inject adapters into all seven projection layers of the Vicuna backbone: `q_proj`, `k_proj`, `v_proj`, `o_proj` (attention), and `gate_proj`, `up_proj`, `down_proj` (MLP feed-forward). This covers both reasoning (attention) and feature transformation (MLP) components. At the default rank $r=16$, the adapters introduce approximately 42 M trainable parameters, or **0.6%** of the 7 B total.

Precision. Our experimental plan called for 4-bit NF4 quantization via QLoRA [5]. During cluster deployment on GMU Hopper, however, `bitsandbytes ≥ 0.43.1` requires `triton.ops.matmul_perf_model`, which is absent from the installed Triton version. We therefore pinned `bitsandbytes==0.41.3` and loaded the base model in `bfloat16`. Since the A100’s 80 GB VRAM accommodates the model at ~ 14 GB, quantization was not required for memory, and `bfloat16` avoids the numerical instabilities of `float16` through its wider exponent range.

3.3. Training Objective

Each training example is formatted using LLaVA’s chat template: the user turn contains an image token and the question; the assistant turn contains the ground-truth answer. Training minimizes the causal language-modeling cross-entropy loss computed only over answer tokens; question and padding tokens are masked with label -100 .

3.4. Training Configuration

All fine-tuning runs share the hyperparameters in Table 1. Gradient checkpointing is enabled to reduce activation memory.

Table 1. Training hyperparameters shared across all experiments.

Hyperparameter	Value
Optimizer	AdamW ($\lambda=0.01$)
Learning rate	2×10^{-4} , cosine decay
Warmup	3% of total steps
Effective batch	16 (size $2 \times$ accum. 8)
Gradient clip	<code>max_norm = 1.0</code>
LoRA rank (default)	$r=16$, $\alpha=32$
Precision	<code>bfloat16 + autocast</code>
Hardware (main)	GMU Hopper A100-SXM4-80 GB
Hardware (sweep)	Kaggle H100-SXM5-80 GB

3.5. Implementation Details

Experiments were conducted on two compute platforms, requiring platform-aware environment management. We implement an environment detection routine that checks for SLURM job identifiers and scratch-space paths to distinguish GMU Hopper (SLURM-managed A100 cluster), Kaggle (H100 notebook environment), and local development machines. Each platform is assigned dedicated directories for HuggingFace model and dataset caches, configured via the `HF_HOME` and `HF_DATASETS_CACHE` environment variables before any library import to ensure deterministic cache routing across job restarts.

On Hopper, pip-installed packages do not persist across SLURM job allocations. Each job therefore begins

with a dependency installation step using pinned package versions before any model code executes. The most critical pin is `bitsandbytes==0.41.3`, identified through iterative debugging of CUDA kernel initialization: later releases introduce a dependency on `triton.ops.matmul_perf_model` that is absent from Hopper’s installed Triton distribution, producing a runtime crash before the model can be loaded.

The LLaVA processor’s tokenizer is configured with right-side padding to maintain consistent attention mask alignment during batched training. Gradient checkpointing is enabled immediately after LoRA injection, trading recomputation of intermediate activations for reduced peak VRAM consumption. A cosine learning-rate schedule with a 3% linear warmup prevents large gradient updates in the early steps when LoRA weight matrices are randomly initialized. All training loss is computed in `bfloat16` via PyTorch’s `autocast` context; the AdamW optimizer maintains full-precision master copies of the LoRA parameters internally, which stabilizes training under mixed-precision.

4. Datasets and Experimental Setup

4.1. Datasets

We use three publicly available Med-VQA benchmarks spanning two medical imaging modalities, summarized in Table 2.

VQA-RAD [1] contains 3,515 clinician-generated QA pairs across 315 radiology images (head, chest, and abdomen CT/MRI/X-ray), split into 1,793 training and 451 test examples. Questions cover organ identification, abnormality detection, and spatial attributes.

PathVQA [7] provides 32,799 QA pairs over 4,998 histopathology images, of which approximately 50% are yes/no questions. We use the HuggingFace split (19,654 train; ~6,700 test).

SLAKE [8] offers 14,028 QA pairs across 642 multi-organ radiology images with semantic knowledge annotations. We use the English-language subset (4,919 train; 1,061 test).

Table 2. Dataset statistics. SLAKE uses English-only subset.

Dataset	Train	Test	Modality
VQA-RAD	1,793	451	CT/MRI/X-ray
PathVQA	19,654	6,700	Histopathology
SLAKE	4,919	1,061	Multi-organ Rad.

4.2. Preprocessing

All images are converted to RGB before processing. Each example is formatted using LLaVA’s processor and chat template, producing input tensors containing pixel

values and tokenized text. A known issue with truncated TIFF files in PathVQA is handled by enabling `PIL.ImageFile.LOAD_TRUNCATED_IMAGES` at import time. We tag every answer as **CLOSED** (exactly “yes” or “no”) or **OPEN** (all other free-form responses) for stratified evaluation. Table 3 reports the resulting distribution across training splits, which directly informs the evaluation dynamics observed in Section 5.

Table 3. Answer-type distribution across training splits.

Dataset	CLOSED (%)	OPEN (%)
VQA-RAD	~57	~43
PathVQA	~50	~50
SLAKE	~40	~60

4.3. Evaluation Metrics

We report three metrics: **closed accuracy**-whether the gold answer appears in the predicted text for yes/no questions; **open recall**-whether any token from the gold answer appears in the prediction for free-form questions; and **overall accuracy**-the combined metric across both types. Open recall is intentionally lenient, appropriate for generative outputs where valid paraphrases should be credited. Both metrics are computed on the full test split unless otherwise noted.

4.4. Experiment Variants

Zero-shot evaluation. LLaVA-1.5-7B is evaluated without any fine-tuning to establish a lower-bound baseline.

Dedicated VQA-RAD fine-tune. Full training split (1,793 rows), 3 epochs, $r=16$, evaluated on the 451-row test set.

Cross-domain transfer matrix. For fair comparison, each training domain is capped at 4,000 shuffled rows, trained for 2 epochs at $r=16$. Every resulting adapter is evaluated on all three test sets ($3 \times 3 = 9$ cells). The cap prevents PathVQA’s 19,654-row split from dominating compute relative to VQA-RAD’s 1,793-row split.

LoRA rank sweep. Conducted on VQA-RAD with $r \in \{4, 8, 16, 32, 64, 128\}$, all other hyperparameters fixed. Executed on Kaggle H100 to exploit the 30 hr/week GPU allocation.

5. Results

5.1. Fine-Tuning Impact on VQA-RAD

Table 4 compares zero-shot LLaVA-1.5-7B against our LoRA fine-tuned model ($r=16$, 3 epochs, full 1,793-row training set, evaluated on all 451 test examples).

Open recall improves the most (+16.43 pp), indicating that the base model already possesses partial yes/no reasoning through its instruction-tuning, whereas free-form clinical answer generation requires explicit domain exposure. The

Table 4. Zero-shot vs. LoRA fine-tune on VQA-RAD test set.

Setting	Closed	Open	Overall
Zero-shot	64.86	41.57	57.04
LoRA $r=16$	75.00	58.00	70.00
Δ	+10.14	+16.43	+12.96

0.6% parameter budget is therefore sufficient to substantially reshape the model’s answer generation behavior without altering its underlying visual representations.

5.2. Cross-Domain Transfer Matrix

Table 5 presents the 3×3 overall accuracy matrix. Rows denote the training domain; columns denote the test domain. All cells use the same compute budget (2 epochs, $\leq 4,000$ training rows, $r=16$).

Table 5. Cross-domain transfer matrix: overall accuracy (%). Bold values are in-domain (diagonal) results.

Train ↓ / Test →	VQA-RAD	PathVQA	SLAKE
VQA-RAD	70.5	38.0	74.0
PathVQA	41.0	54.0	43.5
SLAKE	52.5	37.5	86.5

In-domain performance. SLAKE achieves the highest in-domain accuracy (86.5%), a gain of +32.5 pp over its zero-shot baseline of 54.0%. VQA-RAD reaches 70.5% in-domain. PathVQA’s in-domain accuracy (54.0%) is the lowest, partly due to its high proportion of yes/no questions and the visual complexity of histopathology.

Radiology-to-radiology transfer. The best off-diagonal result is VQA-RAD $\rightarrow$ SLAKE (74.0%), confirming that features learned from one radiology corpus generalize across organ systems. Notably, the reverse direction (SLAKE $\rightarrow$ VQA-RAD) yields only 52.5%-well below the VQA-RAD in-domain score-revealing a strong asymmetry. We attribute this to the clinical specificity of VQA-RAD questions: the compact, focused nature of its training pairs teaches more domain-transferable representations than the broader vocabulary of SLAKE.

Pathology isolation. PathVQA neither sends nor receives effective transfer. Both the PathVQA row and column contain the coldest values in the matrix. Histopathology images are visually distinct from CT/MRI at every level-color distribution, spatial frequency, and semantic content- and the CLIP encoder, pretrained primarily on natural images, lacks

the feature representations necessary for histology understanding. LoRA, which adapts only the language model and not the vision encoder, cannot bridge this gap.

5.3. LoRA Rank Sweep

Table 6 presents the key operating points from sweeping LoRA rank on VQA-RAD across six configurations.

Table 6. LoRA rank sweep on VQA-RAD overall accuracy. Bold marks the Pareto-optimal operating point.

Rank r	Params (M)	Overall (%)
4	10.5	<65
8	21.0	~ 67
16	42.0	~ 70
32	84.0	~ 75
64	168.5	82.0
128	337.0	<82

Performance rises monotonically from $r=4$ to $r=64$, then decreases at $r=128$. The inverted-U pattern indicates a classic capacity–data trade-off: VQA-RAD’s 1,793 training examples cannot support the 337 M adapter parameters introduced at $r=128$, causing overfitting. The Pareto-optimal point is $r=64$ (82.0% overall, 2.4% of total model parameters), where additional rank no longer yields accuracy gains proportional to its parameter cost.

6. Analysis

Why open recall improves more than closed accuracy. Closed questions (yes/no) admit partial coverage from the model’s general instruction-following ability already instilled during LLaVA’s pretraining on web-scale image-text conversations. Open questions, by contrast, require clinical vocabulary-terms such as “consolidation,” “pleural effusion,” and “cortical thinning”- that lie outside LLaVA’s web-pretrained distribution. Fine-tuning on medical QA pairs exposes the language model to these terms in their correct visual context, substantially improving recall of domain-specific answer tokens. This asymmetry suggests that the closed-question ceiling for zero-shot VLMs is higher than the open-question ceiling, and that LoRA most efficiently closes the gap where the performance deficit is largest.

Why VQA-RAD transfers to SLAKE but not vice versa. VQA-RAD questions were authored by clinicians and target precise anatomical and pathological concepts in compact phrasing. Adapters trained on these high-signal pairs learn concise, generalizable medical representations. SLAKE’s broader, knowledge-annotated question set yields adapters that encode a wider but shallower medical vocabulary, which does not compress as effectively into the targeted clinical focus of VQA-RAD. The asymmetry therefore reflects source

dataset specificity more than dataset size: VQA-RAD contains only 1,793 training rows compared to SLAKE’s 4,919, yet it produces the stronger-transferring adapter.

Why pathology does not transfer. The CLIP ViT-L/14 vision encoder was not pretrained on histopathology. Color-based tissue staining (H&E), cellular-scale textures, and the absence of organ-level anatomy mean that CLIP extracts qualitatively different feature vectors for pathology than for radiology. Since our LoRA adapters modify only the language model weights downstream of the frozen encoder, they cannot compensate for a misaligned visual representation space. Future work using a pathology-specific encoder or a fully fine-tuned LLaVA-Med backbone may resolve this gap.

Implications of the answer-type distribution. The CLOSED/OPEN split of each dataset materially shapes evaluation dynamics. VQA-RAD is the most clinically focused, with approximately 57% closed questions testing binary abnormality decisions. PathVQA sits near 50%/50%, making its in-domain score sensitive to yes/no accuracy. SLAKE is the most open-ended at ~60% open questions, which helps explain why its +32.5 pp in-domain gain is the largest in absolute terms: the base model’s open-recall deficit on radiology vocabulary is deep, and fine-tuning on SLAKE’s semantically rich question set closes it effectively. Conversely, a high closed-question ratio provides less gradient signal per training example for improving free-form generation, which limits the cross-domain utility of PathVQA adapters even within the pathology domain itself.

7. Limitations and Future Work

Incomplete CNN+LSTM baseline. A discriminative CNN+LSTM baseline (ResNet-50 image encoder + bidirectional LSTM question encoder + classification head) was implemented but not fully evaluated, as compute resources were prioritized toward the fine-tuning and transfer-matrix experiments. A complete architectural comparison against the VLM pipeline—including accuracy, inference latency, and memory footprint—remains future work.

Per-dataset PathVQA and SLAKE fine-tunes. Only cross-domain runs at a capped training budget (4,000 rows) were completed. Individual full-scale fine-tunes at the natural dataset sizes would provide stronger in-domain upper bounds and a more precise characterization of the rank–accuracy trade-off across modalities.

Permissive open-question metric. Token-level recall credits any prediction that contains a gold-answer word. Clinical validation would require radiologist scoring on a representative sample, providing a more meaningful measure of practical utility beyond token overlap.

Single base model. Experiments are confined to LLaVA-1.5-7B. LLaVA-Med, pretrained on biomedical image-text pairs, may exhibit a different cross-domain transfer pattern, particularly for pathology.

Future directions. (1) *Mixture-of-LoRAs*: train modality-specific adapters (radiology vs. pathology) and gate them at inference time. (2) *Pathology-aware encoder*: replace the frozen CLIP encoder with a domain-specific histopathology foundation model. (3) *Clinical evaluation*: radiologist-scored accuracy on a representative sample beyond token recall.

8. What We Learned

Andres Cabrera. Building the LoRA fine-tuning pipeline end-to-end on a SLURM cluster exposed me to the practical gap between a method paper and a working implementation. Pinning `bitsandbytes` versions, diagnosing Triton kernel incompatibilities, and managing HuggingFace cache across job allocations required careful environment management that is rarely discussed in the literature. Setting up the training loop—cosine scheduling, warmup ratios, gradient clipping, and gradient checkpointing—taught me how each piece interacts under a fixed memory budget. The LoRA rank sweep was particularly instructive: I expected higher rank to monotonically improve accuracy, but the drop at $r=128$ made concrete the intuition that adapter capacity must be matched to training data size. More broadly, the project reinforced that parameter-efficient fine-tuning is a genuine research tool, not merely an engineering convenience, and that controlled ablations are how you distinguish signal from noise in empirical deep learning work.

Mann Mihir Desai. My primary contributions—PathVQA and SLAKE fine-tuning, the cross-domain evaluation harness, error analysis, and visualizations—gave me hands-on experience with the full experimental lifecycle of a multi-dataset study. Writing the evaluation harness required loading multiple LoRA checkpoints sequentially, purging GPU memory between evaluations, and accumulating per-domain metrics without cross-contamination. The transfer matrix results challenged my prior assumptions: I expected dataset size to be the primary driver of transfer quality, but the VQA-RAD → SLAKE asymmetry demonstrated that source specificity matters more than volume. Designing the related work section also deepened my understanding of how the field evolved from discriminative classifiers through self-supervised pretraining to generative VLMs, and clarified exactly what our controlled transfer study adds that prior work does not provide.

9. Conclusion

We have presented a parameter-efficient study of cross-domain transfer and LoRA rank efficiency for medical VQA using LLaVA-1.5-7B. Fine-tuning with as few as 0.6% of model parameters yields a +12.96 percentage-point overall accuracy gain on VQA-RAD, with the largest improvement on open-ended clinical questions where the zero-shot deficit is deepest. A 3×3 cross-domain transfer matrix reveals that radiology domains generalize asymmetrically-with VQA-RAD adapters transferring more effectively to SLAKE than the reverse-while pathology constitutes a visually isolated domain requiring modality-specific representation learning beyond what LoRA alone can provide. A rank sweep identifies $r=64$ as the Pareto-optimal configuration for VQA-RAD, beyond which adapter capacity exceeds what the available training data can support. Together, these findings provide a practical recipe for adapting general-purpose VLMs to specialized medical domains at low cost, and highlight modality-aware adapter design as the most promising direction for extending this work.

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